

TASK 7

Write a JAVA Program to implement the RSA algorithm.

Aim: To implement RSA Algorithm.

Program:

```
import java.math.*;
import java.util.*;
//RSA
public class Task7 {
    static Scanner s = new Scanner(System.in);
    private static BigInteger p, q, N, phi, e, d;
    private static int bitlength = 1024;
    private static Random r;
    private static void Setter() {
        r = new Random();
        p = BigInteger.probablePrime(bitlength, r);
        q = BigInteger.probablePrime(bitlength, r);
        N = p.multiply(q);
        phi = p.subtract(BigInteger.ONE).multiply(q.subtract(BigInteger.ONE));
        e = BigInteger.probablePrime(bitlength / 2, r);

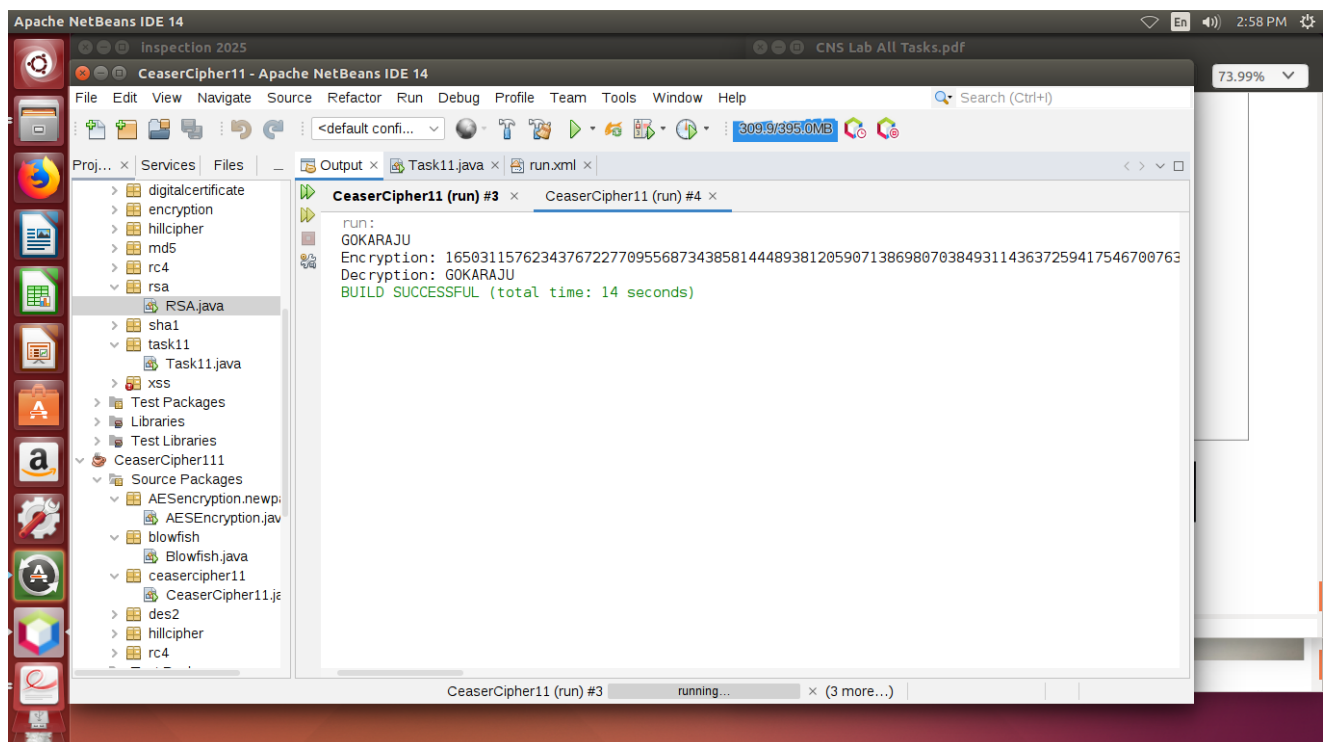
        while (phi.gcd(e).compareTo(BigInteger.ONE) > 0 && e.compareTo(phi) < 0) {
            e = e.add(BigInteger.ONE);
        }
        d = e.modInverse(phi);
    }
    private static byte[] encrypt(String data) {
        return (new BigInteger(data.getBytes())).modPow(e, N).toByteArray();
    }
    private static byte[] decrypt(byte[] encryptedData) {
        return (new BigInteger(encryptedData)).modPow(d, N).toByteArray();
    }
    public static void main(String[] args) { String data = s.nextLine();
        Setter();
        // Encryption
```

```

byte[] encrypted = encrypt(data);
System.out.println("Encryption: " + new BigInteger(encrypted).toString());
// Decryption
System.out.println("Decryption: " + new String(decrypt(encrypted)));
}
}

```

Output:



VIVA QUESTIONS

1. What is the RSA algorithm, and how does it work?
2. How are the public and private keys generated in RSA?
3. What are the main advantages and disadvantages of using RSA for encryption?
4. How does RSA encryption compare to symmetric encryption algorithms like AES in terms of performance and security?